AZ-305 Agenda

Module 01 Design a governance solution 

Module 02 Design a compute solution

Module 03 Design a non-relational data storage solution

Module 04 Design a data storage solution for relational data

Module 05 Design a data integration solution

Module 06 Design an application architecture solution

Module 07 Design Authentication and Authorization Solutions

Module 08 Design a solution to log and monitor Azure resources

Module 09 Design a network infrastructure solution

Module 10 Design a business continuity solution

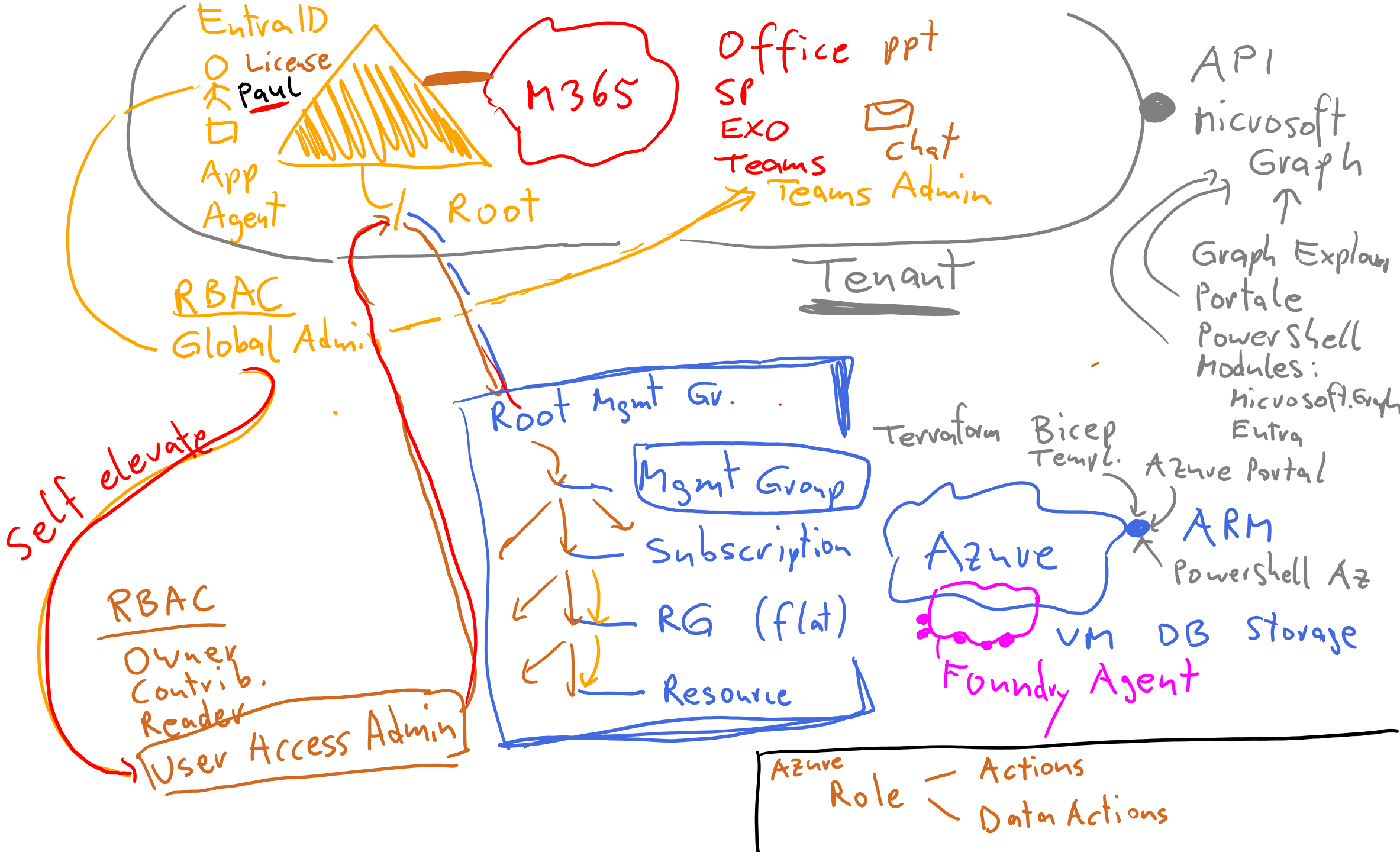
Module 11 Design a migration solution

Design a governance solution



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Entra ID
License
Paul
App
Agent
M365
Root
Office ppt
SP
EXO
Teams
Chat
Teams Admin
Tenant
API
Microsoft
Graph
Graph Explorer
Portals
PowerShell
Modules:
Microsoft.Graph
Entra
RBAC
Global Admin
Self elevate
Root Mgmt Gr.
Mgmt Group
Subscription
RG (flat)
Resource
Terraform
Bicep
Templ.
Azure Portal
ARM
PowerShell Az
Azure
VM
DB
Storage
Foundry Agent
RBAC
Owner
Contrib.
Reader
User Access Admin
Azure
Role
Actions
Data Actions

Learning Objectives

- Design for governance
- Design for management groups
- Design for Azure subscriptions
- Design for resource groups
- Design for resource tagging
- Design for Azure Policy and RBAC
- Design for Azure Landing Zones
- Case study
- Learning recap

AZ-305: Design Identity, Governance, and Monitoring Solutions (25-30%)

Design Governance

- Recommend a structure for management groups, subscriptions, and resource groups, and a strategy for resource tagging
- Recommend a solution for managing compliance
- Recommend a solution for identity governance

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Design for governance

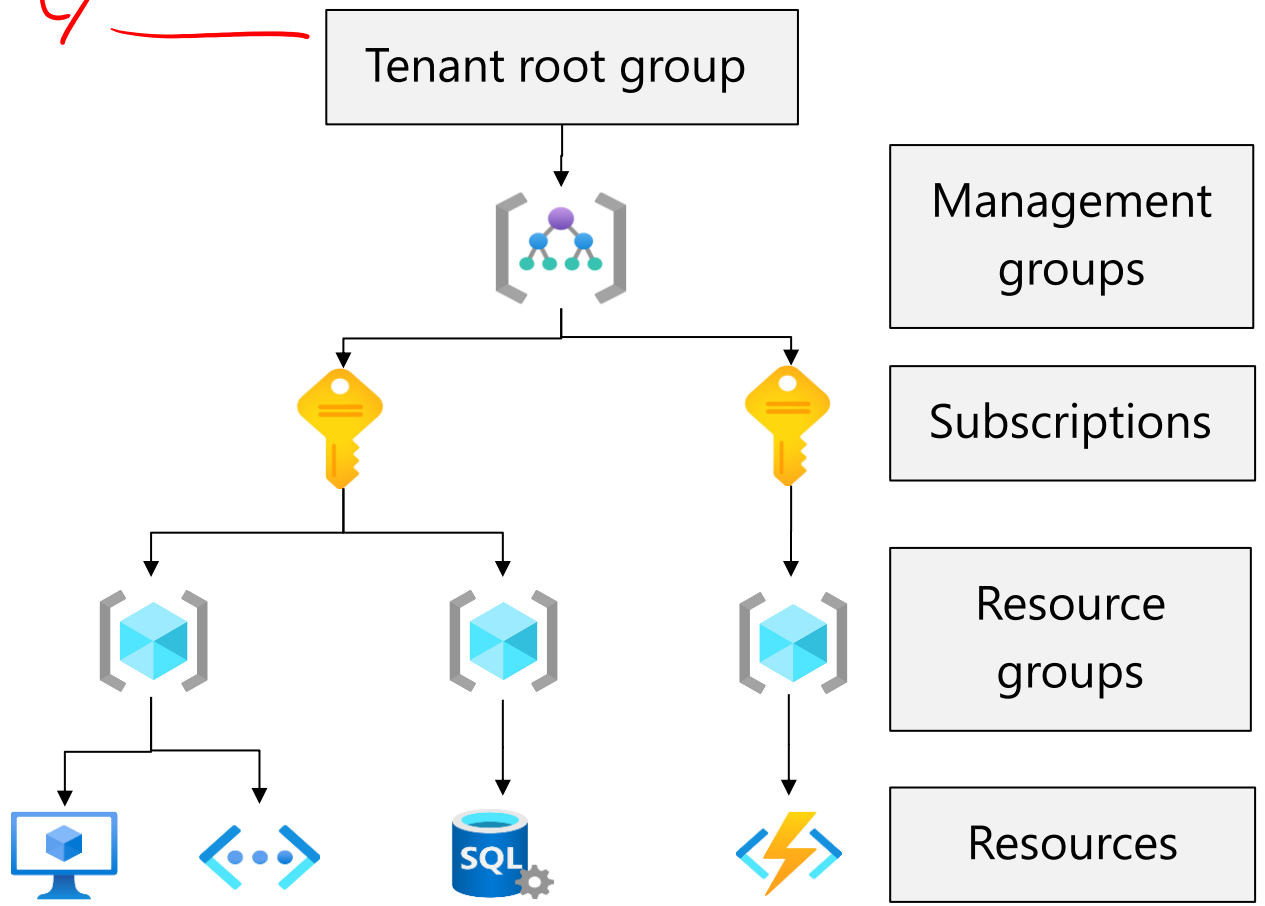


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Govern resources in Azure

Governance provides mechanisms and processes to maintain control over your applications and resources in Azure.

- Determine your requirements, plan your initiatives, and set strategic priorities
- Plan for governance at every level
  - Management groups
  - Subscriptions
  - Resource groups
  - Resources



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Design for management groups

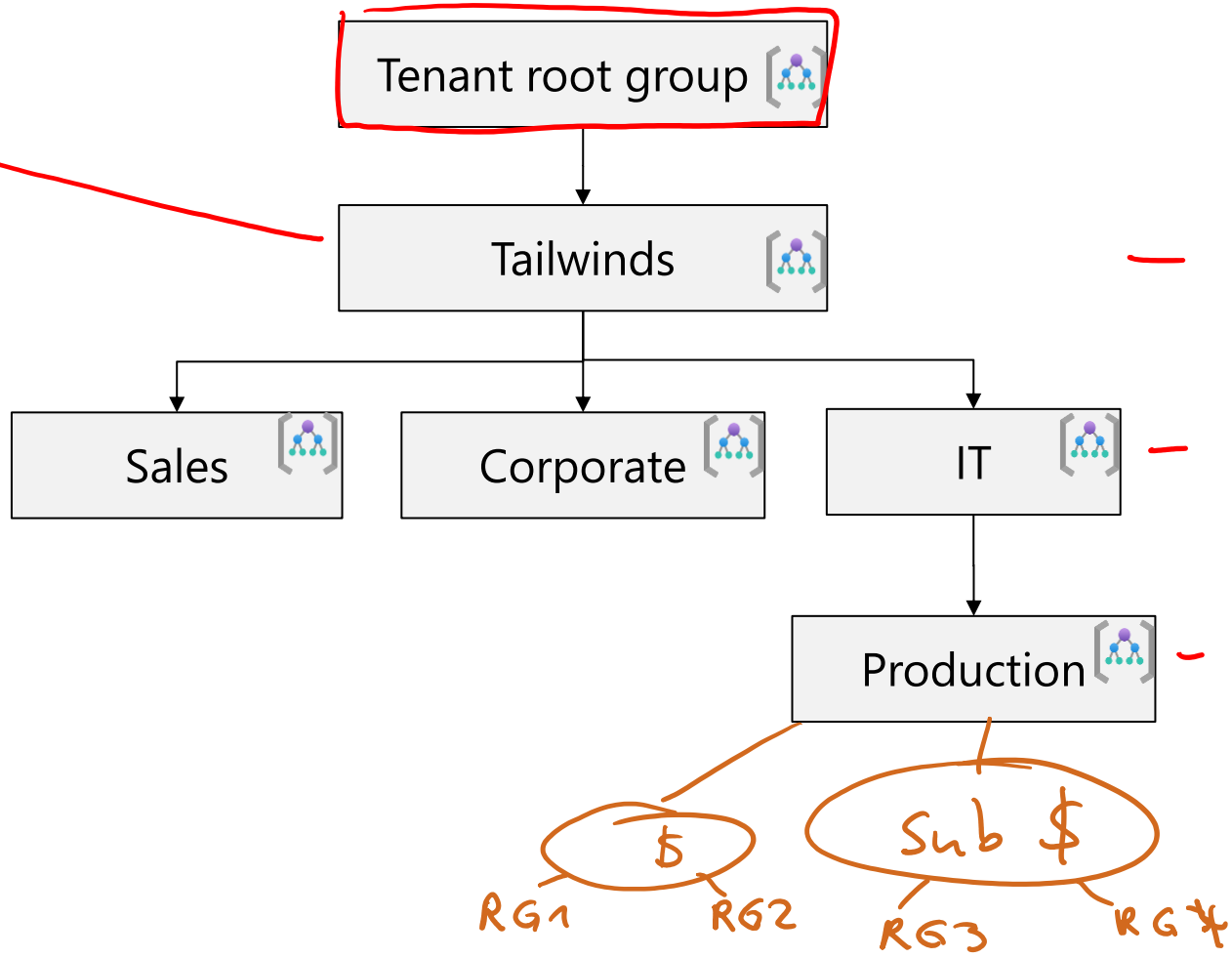


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Plan your management groups

Management groups manage access, policy, and compliance for multiple subscriptions.

- Keep the management group hierarchy reasonably flat
- Consider a top-level management group
- Consider an organizational or departmental structure
- Consider a geographical structure
- Consider a production management group
- Consider a sandbox management group
- Consider isolating sensitive information in a separate management group



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Design for Azure subscriptions



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Designing for multiple subscriptions

Azure subscription are logical containers for management and billing.



Align your subscriptions with business needs and priorities – consider billing and cost reporting



Consider subscription scale limits – specialized workloads, IoT, SAP



Consider administrative management – centralized or decentralized



Consider a dedicated shared services subscription – common services everyone shares



Group subscriptions together under management groups – apply common policies and role assignments.

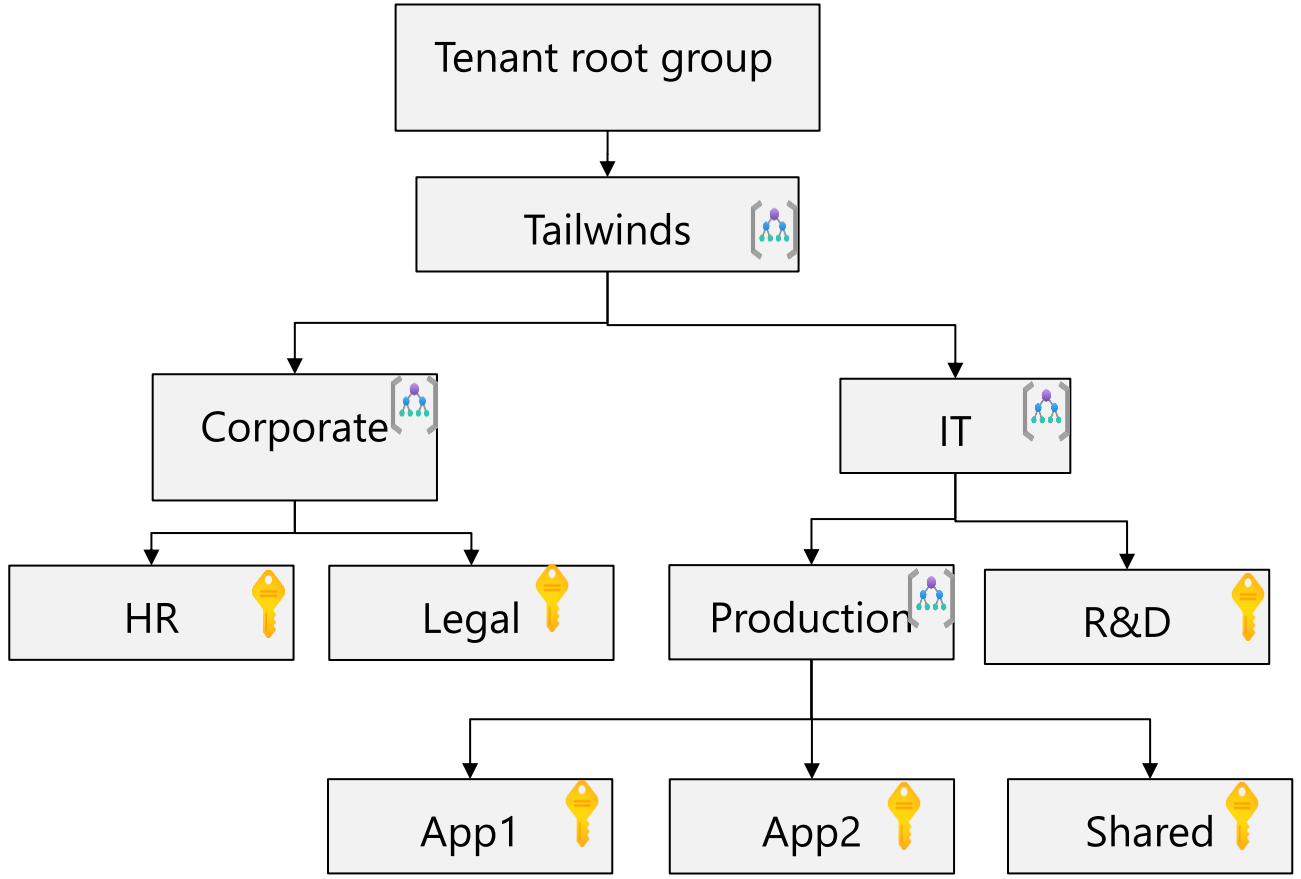


Make subscription owners aware of their roles and responsibilities

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When to use subscriptions - example

- Secure workloads that require additional policies and role-based access control to achieve compliance
- Specialized workloads and the need to scale outside the subscription limits
- Manage and track costs for your organizational structure
- Identify different environments such as development, test, and production that are often isolated from a management perspective



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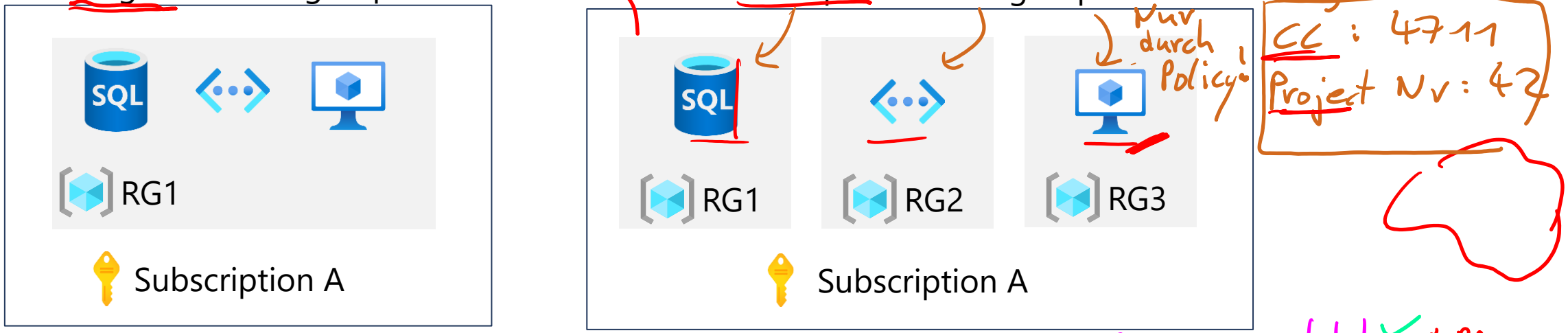
Design for resource groups



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Plan your resource groups

A resource group is a container that holds related resources for an Azure solution.



- Group resources that share the same life cycle
- Group by type, app, department, location, or billing
- Apply RBAC and policies to a group of resources
- Use resource locks to protect individual resources from deletion or change

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Design for resource tagging

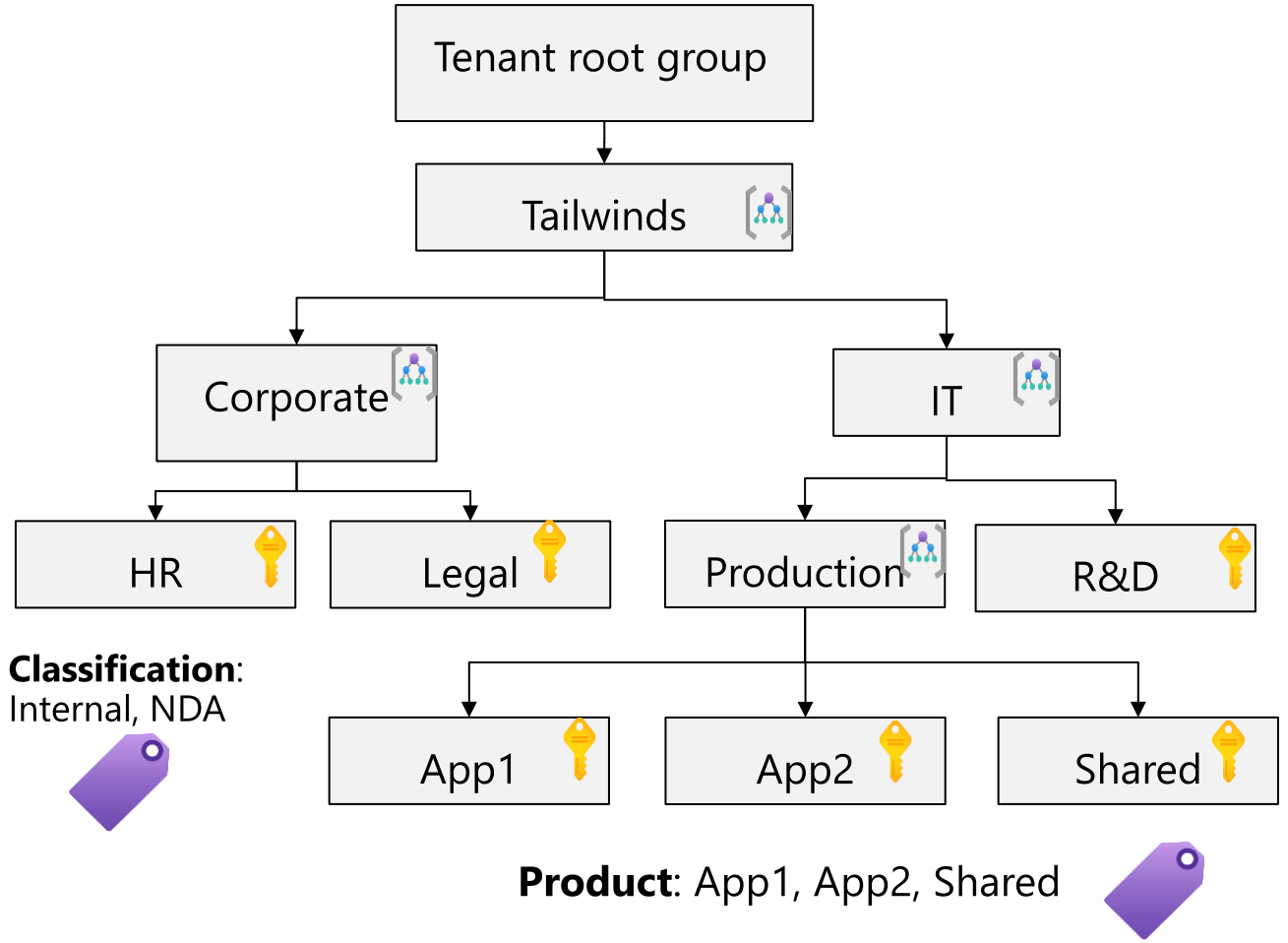


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Plan your resource tagging

Resource tagging can be business-aligned or IT-aligned

- Consider your organization's taxonomy
- Determine the reason for the tagging - functional, classification, accounting, partnership, or purpose
- Start with a few tags (mission-critical resources) and then scale out
- Policies could be used to apply tags and enforce tagging rules and conventions - mimic inheritance



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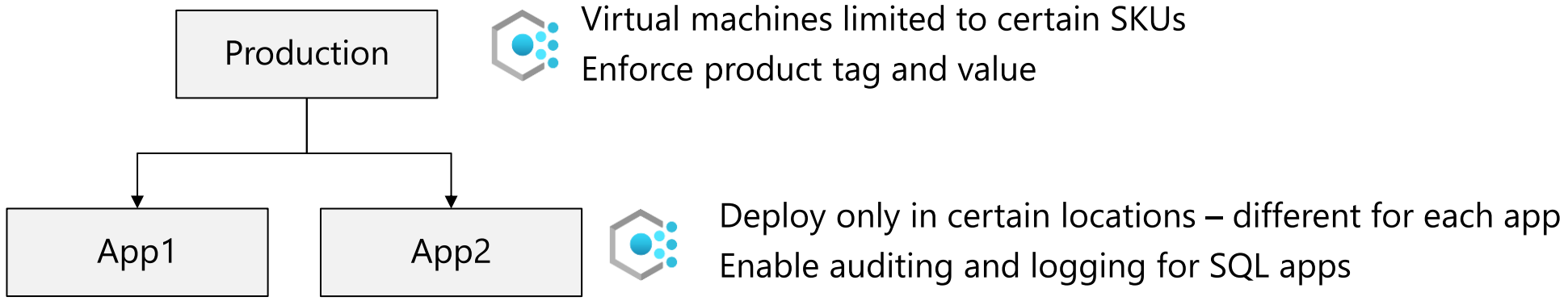
Design for Azure Policy and RBAC



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When to use Azure Policy

Azure Policy helps to enforce organizational standards and to assess compliance at-scale.



- Large number of built-in policies and you can create custom policies

Examples

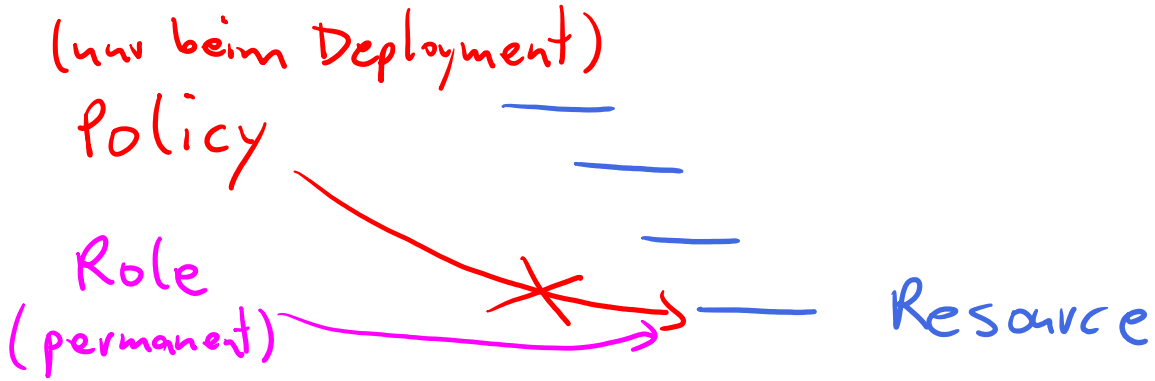
- Allow only certain virtual machines sizes for your project
- Ensure all resources are correctly tagged – if not, apply the tag
- Recommend system updates on your servers
- Enable multifactor authentication for all subscription accounts

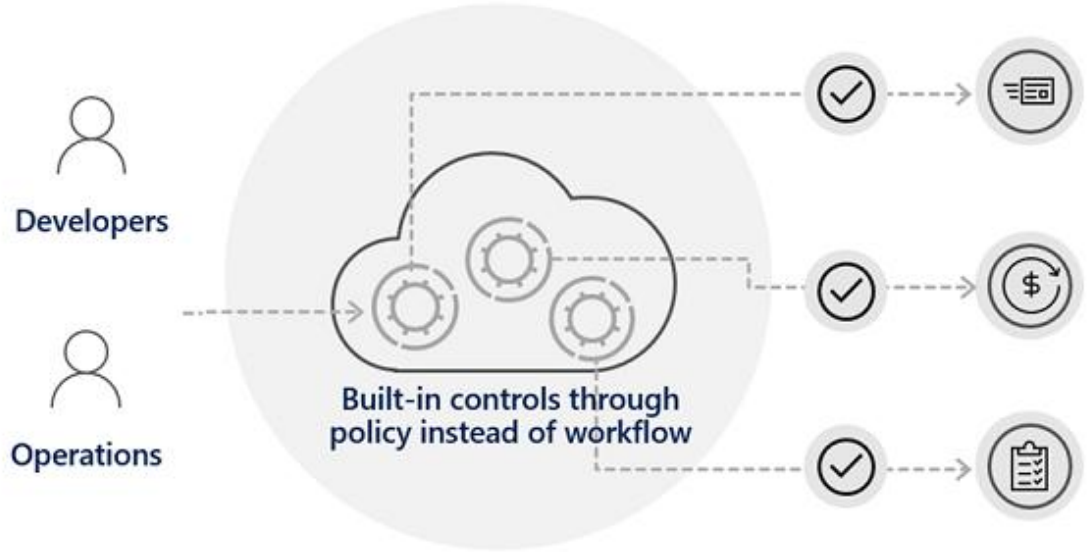
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Considerations for Azure Policy

- Apply policy at the highest scope possible
- Know when policies are evaluated
- Decide what to do if a resource is non-compliant
- Consider when to automatically remediate non-compliant resources
- Use the Azure policy compliance dashboard for auditing and review
- Effectively combine Azure policy with RBAC (RBAC on next slide, combination on slide after)





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Design for Azure role-based access control (RBAC)

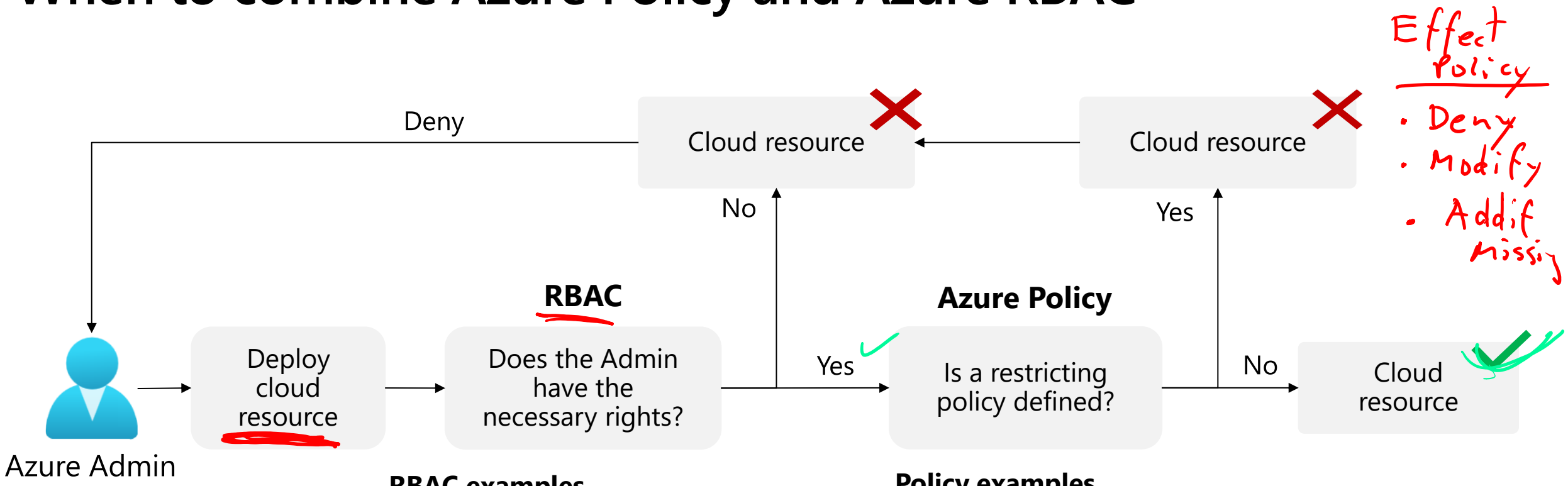
Azure RBAC allows you to grant access to Azure resources that you control.

- Only grant users the access they need
- Assign at the highest scope level that meets the requirements
- Assign roles to groups, not users
- Know when to create a custom role
- Consider what happens if you have overlapping role assignments



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When to combine Azure Policy and Azure RBAC



RBAC examples

- Does the Admin have the right to deploy?
- Does the Admin have the right to deploy this resource type?
- Does the Admin have the right to deploy this resource group?

Policy examples

- Is the region restricted?
- Is the resource type restricted?
- Should a tag be applied?

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Design for Landing Zones

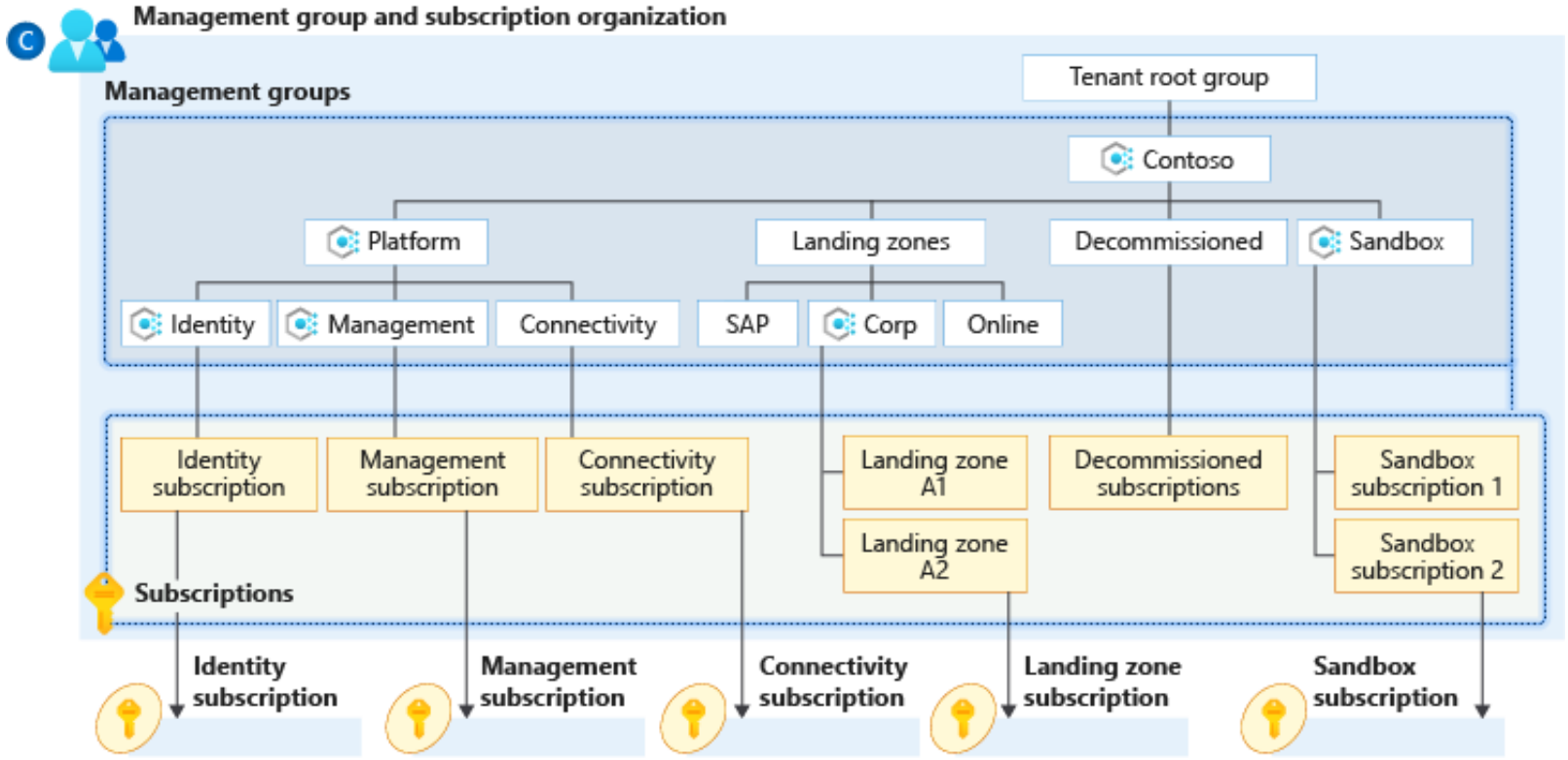


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Implement Landing Zones

A landing zone provides an infrastructure environment for hosting your workloads.

- Implements key foundational principles of governance, security, networking, management, and identity
- Pre-provisions the environment through code
- Good for both migrations and green field situations
- You can transition existing architectures
- Part of the Cloud Adoption Framework Ready phase



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Case Studies and Review



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Case Study – Cost and accounting

- Tailwind Traders has two main business units that handle Apparel, and Sporting Goods.
- Each of the business units consist of three departments: Product Development, Marketing, and Sales.
- Each business unit and subunit will be responsible for tracking their Azure spend.
- The Enterprise IT team will be responsible for providing company-wide Azure cost reporting.

- What are different ways Tailwind Traders could organize their subscriptions and management groups. Which would be the best to meet their requirements?
- Design two alternative hierarchies and explain your decision-making process.

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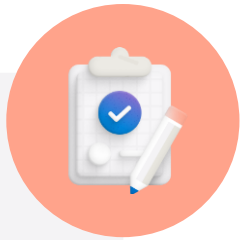
Case study – New development project

- The company has a new development project for customer feedback.
- The CFO wants to ensure all costs associated with the project are captured.
- For the testing phase workloads should be hosted on lower cost virtual machines.
- The virtual machines should be named to indicate they are part of the project.
- Any instances of non-compliance with resource consistency rules should be automatically identified.

- What are the different way Tailwind Traders could track costs for the new development project?
- How are you ensuring compliance with the requirements for virtual machine sizing and naming?
- Propose at least two ways of meeting the requirements. Explain your final decision.

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Learning Recap – Governance solutions



Check your knowledge questions and review

Module references

- Control and organize Azure resources with Azure Resource Manager
- Introduction to Microsoft Azure: Describe cloud concepts
- Introduction to Microsoft Azure: Describe Azure management and governance
- Introduction to the Microsoft Azure Well-Architected Framework

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End of presentation



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